

PostgreSQL on AWS – Performance Playbook (Advanced)

Balanced 50/50 read–write workload guidance for AWS RDS and Aurora PostgreSQL. Includes EXPLAIN ANALYZE examples, query archetypes, and outage response.

1. Real EXPLAIN ANALYZE Before / After

Before: Sequential scan due to missing composite index

```
EXPLAIN ANALYZE SELECT * FROM orders WHERE customer_id = 12345 ORDER BY created_at DESC LIMIT 10; Seq Scan on orders Execution Time: 842 ms
```

After: Index eliminates scan and sort

```
Index Scan using idx_orders_customer_created Execution Time: 1.4 ms
```

2. Query Archetype Library

Type	Pattern	Primary Optimizations
OLTP	Small indexed lookups	Composite indexes, short TXs
Reporting	Aggregates, joins	Materialized views, replicas
Time-Series	Append + time filters	Partitioning, BRIN

3. Outage Performance Quick Sheet

- Identify impact: latency vs errors
- Check pg_stat_activity and locks
- Inspect pg_stat_statements top queries
- Run EXPLAIN ANALYZE on worst offender
- Check CloudWatch (CPU, IOPS, latency)
- Apply lowest-risk fix first
- Document findings

4. AWS Balanced Workload (50/50) Tuning

- Avoid excess indexes—writes pay the cost
- Use Aurora readers or RDS replicas for reports
- Tune autovacuum for steady churn
- Set statement_timeout defensively
- Watch IOPS before CPU saturation
- Use Performance Insights as primary signal